

Spoken Tutorial on Understanding APIs

Module	AI Essentials
Episode	2. Understanding APIs
Learning Objective	At the end of this tutorial learner will be able to 1. Define what an API is. 2. Explain the role of an API as a connector. 3. Identify how prompts are sent via APIs. 4. Apply API keys for authentication.
Approx. Duration	3-4 min
Outline	• What an API is and why it matters • Prompts as inputs sent through APIs • API keys for authentication and access control • How applications send requests and receive AI-generated responses
Meta Tags	API, Application Programming Interface, AI, requests, responses, authentication, API key, prompts, connector, software development
Pre-requisite Tutorial	Episode 1: Python basics for API

Script

Visual Cue	Narration
Title Slide	Welcome to this Spoken Tutorial on Understanding APIs .
Learning Objectives Slide	In this tutorial, you will learn to, Define what an API is. Explain the role of an API as a connector. Identify how prompts are sent via APIs . Apply API keys for authentication.
System Requirements Slide	Here I am using a browser on a computer or mobile.
Pre-requisite Slide	To follow this tutorial, you should have completed Episode 1: Python basics for API .
Illustration of a waiter connecting a customer (app) to the kitchen (server) with digital elements	Have you ever wondered how different apps talk to each other? An API , or Application Programming Interface , is a set of rules. It lets different software applications communicate smoothly. Think of an API like a waiter in a restaurant. You give your order, or request , to the waiter. The waiter takes it to the kitchen, which is the server. Then, they bring back your food, which is the response . This digital waiter helps apps work together effortlessly.

Illustration of a person ordering food from a menu with thought bubbles showing a prompt	Now, how do you tell an AI what you want? Prompts are your specific instructions or questions. You send these prompts to an AI model using an API. Imagine you are at a restaurant, looking at the menu. A prompt is like ordering a specific dish from that menu. You tell the kitchen exactly what you desire. This ensures the AI understands your request clearly.
Illustration of a VIP pass or a key unlocking a digital gate	What stops just anyone from using an API? API keys are special credentials. They are used for authentication and authorization. This ensures only authorized users can access the API. Think of an API key as a VIP pass. It's like a reservation number for an exclusive service. This pass grants you access to use the API. It keeps your interactions with AI secure.
Screen close-up: user typing 'Tell me something...'	Let us see an example of sending a request. Here is a poorly constructed request to an AI API. Notice the lack of clarity and necessary parameters. This prompt is too vague for the AI to understand. What kind of response do you expect from this?
Screen close-up: user typing 'Generate a Python function for API authentication...'	Now, let us try an improved approach. This request is well-formed and clear. It includes proper parameters and an API key. The instructions are specific for the AI model. See how using an API key enhances security. This clarity ensures a better AI generated response.
Side-by-side: vague prompt output vs improved prompt output	Let's compare the vague and improved requests. On the left, you see the vague prompt. On the right, notice the improved, specific prompt. Clear prompting and proper API usage are vital. They directly impact the quality of AI responses. See the difference in expected output.
Summary Slide	Let us summarize what we learned. We defined what APIs are and their function. We saw how APIs connect different applications. We understood the role of prompts and API keys. Finally, we explored sending requests to AI models.
Assignment Slide	Now as an assignment, Create a program to send a prompt to a public API. Then, print the received response. Experiment with different kinds of prompts. Compare the results.
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